csbewma: An R Package for the Cumulative Standardized Binomial EWMA Control Chart for Multiple Stream Processes

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Abstract Monitoring binomial proportions across multiple independent streams arises routinely in statistical process control, with applications in manufacturing, healthcare, cybersecurity, finance, and environmental monitoring. Exponentially weighted moving average (EWMA) charts are attractive in this setting for their sensitivity to small and moderate shifts, but existing EWMA implementations for binomial data rely on asymptotic variance approximations that are inaccurate during the early phase of monitoring, when the sample history is short. This paper introduces csbewma, an R package implementing the Cumulative Standardized Binomial EWMA (CSB-EWMA) control chart, which uses an exact, closed-form, time-varying variance for the EWMA statistic under a binomial data-generating process. This enables adaptive control limits that are valid from the first observation rather than only asymptotically. Beyond the charting engine, csbewma provides a post-hoc identification framework for determining which stream or streams triggered a signal, using exact binomial p-values with Bonferroni, Holm, or Benjamini-Hochberg multiplicity correction, along with a set of ggplot2-based plotting functions for control charts and diagnostic displays. The package's functions and S3 class design are described, its usage is illustrated with real console output and control charts, and the simulation evidence behind its default settings is summarized. csbewma is distributed under the MIT license and is available from the Comprehensive R Archive Network.

0.1 Introduction

Statistical process control is a foundational methodology for monitoring, controlling, and improving processes across manufacturing, healthcare, finance, and service industries (Montgomery, 2019). Since Shewhart introduced the control chart in the 1920s (Shewhart, 1931), the central task of statistical process control has been to separate common cause variation, which is inherent to a stable process, from special cause variation that signals a change requiring intervention. Traditional Shewhart charts are effective for detecting large shifts but are comparatively insensitive to small and moderate process changes. This motivated memory-type alternatives such as the cumulative sum (CUSUM) chart (Page, 1954) and the exponentially weighted moving average (EWMA) chart (Roberts, 1959), which incorporate the full process history through exponential weighting and are prized for their sensitivity to small, persistent shifts (Lucas and Saccucci, 1990).

A growing class of applications requires monitoring several parallel data streams at once rather than a single process, so-called multiple stream processes, which originated in manufacturing settings such as multi-head filling machines and multi-spindle operations (Boyd, 1950). A more recent line of work pursues distribution-free multiple stream process monitoring by dichotomizing each stream's observations relative to an in-control median, reducing the monitoring problem to tracking a binomial proportion. The nonparametric extended median test CUSUM chart of (Brown and Schaffer, 2021) is the primary contribution of this kind, though it focuses on detecting comparatively large shifts and relies on asymptotic properties that are less suitable for small-sample, early-phase monitoring.

Applying the more sensitive EWMA framework to binomial multiple stream data exposes a genuine theoretical gap. Existing EWMA charts for binomial or count data (Gan, 1990, 1993; Anastasopoulou and Rakitzis, 2022) rely on an asymptotic variance approximation that does not capture the time-varying nature of the true variance during the initial monitoring phase, when data are limited. Because most implementations then use constant control limits derived from this asymptotic variance, they either signal falsely too often or fail to exploit the increasing precision that accrues with each new observation.

This gap is closed by deriving the exact, closed-form, time-varying mean and variance of the EWMA statistic for cumulative binomial counts observed across k independent streams (Muritala et al., 2026a,b). The resulting Cumulative Standardized Binomial EWMA (CSB-EWMA) chart uses these exact moments to construct adaptive control limits that are statistically valid from the very first observation, rather than only in the large-sample limit, while retaining the distribution-free property of the underlying dichotomization approach.

This paper documents the csbewma package, the R implementation of that methodology, with an emphasis on how to use it. The Statistical methodology section summarizes the theory so that the package's arguments and defaults make sense. The csbewma package section describes the package's design and functions. The Illustrative use section is the core usage illustration: real code and the real console output and control charts it produces when this document is rendered. The Chart parameter selection section summarizes the simulation evidence behind the recommended λ and L settings. The Post-hoc identification section documents the post-hoc identification methodology, its rationale, its simulated performance, and a worked illustration of its early-signal limits. The Limitations section discusses the method's limitations, and the final section concludes.

0.2 Statistical methodology

This section summarizes, at a level sufficient to understand the package's arguments, the statistical derivation underlying the CSB-EWMA chart. Full derivations and proofs appear in (Muritala et al., 2026a) and Chapter 3 of (Muritala, 2026).

Process framework

The chart monitors k mutually independent data streams. Within a given stream, observations collected at successive sampling times are also assumed mutually independent, and the process must be measurable on at least an ordinal scale (Brown and Schaffer, 2021). Let y_{ij} denote the jth observation from stream i, $i = 1, \ldots, k$, which has an unknown probability distribution but a known, stable in-control median. A binary indicator x_{ij} is defined as 1 if y_{ij} exceeds that median and 0 otherwise. Provided stream i's median remains stable, x_{ij} follows a $\text{Binomial}(n = 1, p_0 = 0.5)$ distribution. Because the streams, and within a time point the observations across streams, are independent, the column total C_j (the sum of x_{ij} over streams $i = 1, \ldots, k$) follows $\text{Binomial}(n = k, p_0 = 0.5)$, and the cumulative count of exceedances through time t, Q_t (the running sum of C_j), follows $\text{Binomial}(n = kt, p_0 = 0.5)$. The standardized statistic is:

$$Q_t \sim \text{Binomial}(kt,\, p_0), \qquad W_t = \frac{Q_t - tkp_0}{\sqrt{tk\, p_0(1 - p_0)}}$$

with mean 0 and variance 1 by construction.

EWMA recursion and its exact moments

The EWMA statistic follows the recursion below, with $r_0 = 0$ in the package and smoothing parameter $0 < \lambda \leq 1$:

$$r_t = \lambda W_t + (1 - \lambda) r_{t-1}, \qquad r_0 = 0, \qquad 0 < \lambda \leq 1$$

(Muritala et al., 2026a) show that $E[r_t] = (1 - \lambda)^t r_0$, which is exactly zero in the package's implementation since $r_0 = 0$. The exact variance follows from the covariance structure $\text{Cov}(W_i, W_j) = \sqrt{\min(i,j)}/\sqrt{\max(i,j)}$ and is given, for finite t, by:

$$\text{Var}(r_t) \;=\; \lambda^2 \left[2 \sum_{j=1}^{t-1} \sum_{i=j+1}^{t} (1 - \lambda)^{2t-i-j} \sqrt{\frac{j}{i}} \;+\; \sum_{i=1}^{t} (1 - \lambda)^{2t-2i} \right]$$

This is precisely what `var_rt_exact_single()` computes, as described in the Exact variance computation section below.

Asymptotic behavior and adaptive control limits

As t grows large, $\text{Var}(r_t)$ approaches 1 (Muritala et al., 2026a, Theorem 3). The package therefore uses time-varying control limits, where L is the control limit multiplier:

$$UCL_t = L\sqrt{\text{Var}(r_t)}, \qquad LCL_t = -L\sqrt{\text{Var}(r_t)}$$

These limits are narrow near $t = 1$ and widen toward $\pm L$ as t grows, which is the mechanism by which the chart avoids both the elevated early-phase false alarm risk and the loss of early-phase sensitivity that occur when a chart instead uses constant, asymptotic limits from the outset.

Table 1: Validation of the exact EWMA mean and variance against Monte Carlo simulation (k = 10, p0 = 0.5, lambda = 0.2, 10,000 replications). Source: Muritala et al. (2026a, Table 4.1); Muritala (2026, Table 3.2).

Time	Theoretical Mean	Simulated Mean	Theoretical Var.	Simulated Var.	Rel. Bias (Var.)
10	0	-0.005540	0.6369	0.6385	0.26%
50	0	-0.005230	0.9512	0.9359	-1.61%
100	0	-0.001950	0.9768	0.9758	-0.10%
500	0	-0.000606	0.9955	0.9967	0.12%
1000	0	-0.003090	0.9978	0.9996	0.19%

Monte Carlo validation

(Muritala et al., 2026a) and Chapter 3 of (Muritala, 2026) validate the closed-form expressions against 10,000 Monte Carlo replications under the in-control process, with $k = 10$ streams, $p_0 = 0.5$, and $\lambda = 0.2$ (R version 4.3.2; R Core Team (2023)). Table 1 reproduces the reported validation metrics; these are cited results from that Monte Carlo study, not values recomputed by this document (a 10,000-replication study is outside the R Journal's ten-minute knit budget), so the table below is built from a small data frame of those reported numbers rather than from a live simulation.

The simulated means are effectively zero throughout, with a reported root mean square bias of 2.89×10^{-3}. The exact variance formula tracks the simulated variance closely at every reported time point, with relative bias below 2% throughout, converging to about 0.19% by $t = 1000$. The variance reaches 99% of its asymptotic value by $t = 227$, which is the basis for the package's default `converge_t = 500` cutoff described below.

0.3 The csbewma package

The csbewma package is distributed on CRAN as version 1.0.1 (published 2026-06-05; DOI 10.32614/CRAN.package.csbewma). Per its DESCRIPTION file, it requires R (>= 3.5), imports ggplot2, stats, and patchwork, suggests testthat (>= 3.0.0), and is distributed under the MIT license. This paper documents eleven exported functions, plus the `print` and `plot` S3 methods for class `csb_ewma`, grouped by role in Table 2.

Table 2: Exported functions and S3 methods of csbewma 1.0.1, by functional group.

Group	Function(s)	Purpose
Exact variance / control limits	var_rt_exact_single(), precompute_variance()	Compute Var(r_t) exactly at a single t, or precompute a full variance vector with asymptotic fallback beyond converge_t.
Charting engine	run_csb_ewma(), csb_ewma(), print.csb_ewma()	Low-level chart algorithm on a binary matrix; user-facing wrapper handling variance precomputation and post-hoc identification; S3 print method.
Post-hoc identification	calculate_pvalues(), apply_multiple_testing(), identify_ooc(), flagged_streams_summary()	Exact binomial p-values per stream; Bonferroni, Holm, BH, or raw multiplicity correction via p.adjust(); combined stream-level identification; summary of flagged streams.
Visualization	plot.csb_ewma(), plot_csb_ewma_direct(), plot_flagged_streams(), plot_chart_with_flagged()	ggplot2 control chart with signal annotation; direct, non-S3 equivalent; bar chart of stream p-values; patchwork dashboard combining the chart and the bar chart.

The Illustrative use section below demonstrates the resulting workflow with real package output.

Exact variance computation

`var_rt_exact_single(lambda, t)` evaluates the closed-form expression above for a single time point, returning 0 when $t = 0$. The function accumulates two pieces exactly as in the derivation: an

Table 3: Required input structure for csb_ewma(): a k stream by t sample recoded binary matrix, with column totals C_j.

Stream	j = 1	j = 2	...	j = t
1	x_11	x_12	...	x_1t
2	x_21	x_22	...	x_2t
...	...	...	...	...
k	x_k1	x_k2	...	x_kt
Column totals	C_1	C_2	...	C_t

off-diagonal contribution, summed for $j = 1, \ldots, t-1$ using vectorized weights $(1-\lambda)^{2t-j-i}$ for i running from $j+1$ to t and the covariance terms $\sqrt{j/i}$, doubled for the symmetric triangle, and a diagonal contribution $(1-\lambda)^{2t-2i}$ summed over $i = 1, \ldots, t$. The total is multiplied by λ^2. Because this double summation costs $O(t^2)$ if evaluated for every t up to a long monitoring horizon, `precompute_variance(lambda, max_t, converge_t = 500)` instead computes the exact variance only for $t = 1, \ldots, \min(\texttt{converge\_t}, \texttt{max\_t})$ and fills the remaining entries, for t greater than `converge_t`, with the asymptotic value 1. The default `converge_t = 500` is set well above the empirical 99% convergence point of $t = 227$ established above; the function's own documentation describes it as safe and efficient. The resulting numeric vector, one entry per time point, is the `var_cache` consumed by `run_csb_ewma()`.

The charting functions

`run_csb_ewma(bin_matrix, lambda, L, var_cache, max_time = NULL, p0 = 0.5)` is the core algorithm, implemented, in the words of its own documentation, exactly as in the original simulation code. After confirming that `bin_matrix` contains only 0s and 1s, that $0 < \lambda \leq 1$, and that $L > 0$, it sets $k = \texttt{nrow(bin\_matrix)}$, $\mu_0 = kp_0$, and $\sigma_0^2 = kp_0(1-p_0)$. It then steps through $t = 1, \ldots, \texttt{max\_time}$, at each step summing the binary indicators across streams for that time point (C_t), accumulating the running total (`cum_sum`), forming the standardized statistic $W_t = (\texttt{cum\_sum} - \mu_0 t)/\sqrt{t\sigma_0^2}$, updating the EWMA r_t, and comparing r_t against the exact control limits above. The loop stops and records `signal_time` the first time r_t exceeds either limit. If no signal occurs by `max_time`, the function issues a warning and sets `T_sig` to `max_time`. It returns an object of S3 class `csb_ewma` holding `signal_time`, `signal_detected`, `T_sig`, the `r_history`, `UCL_history`, and `LCL_history` vectors truncated to `T_sig`, the truncated `bin_matrix`, per-stream successes, k, λ, L, and p_0.

`csb_ewma(data, lambda, L, p0 = 0.5, max_time = NULL, posthoc_method = "BH", alpha = 0.05)` is the function most users call directly. It validates that `data` is a matrix with $0 < \lambda \leq 1$ and $L > 0$, precomputes the variance vector with `precompute_variance()`, and runs `run_csb_ewma()`. If a signal is detected, it automatically performs post-hoc identification by calling `identify_ooc()` on the truncated binary matrix with the requested `posthoc_method` (Benjamini-Hochberg, `"BH"`, by default) and `alpha`, attaching the resulting data frame as `$flagged` along with `$posthoc_method` and `$alpha`. It also prints short status messages as it runs, for example confirming that the input is binary, reporting the signal time, and reporting whether any streams were flagged, which the Illustrative use section reproduces live. `print.csb_ewma()` then displays λ, L, p_0, and k, whether and when a signal occurred, and, when post-hoc results are attached, the post-hoc method, alpha, the number of flagged streams, and their stream numbers if any were flagged.

`csb_ewma()` takes `data` as an already binary 0/1 matrix; it does not accept a `distribution` argument, as a direct console session confirms. The `data` argument is `bin_matrix`, a k by t matrix with one row per stream and one column per sampling time, entries $x_{ij} \in \{0, 1\}$, formed exactly as in the process framework described above: $x_{ij} = 1$ if the jth observation from stream i exceeds that stream's known, stable in-control median (or otherwise indicates the monitored event), and 0 otherwise. Table 3 lays out this structure explicitly.

Because it is assumed that, for a given sampling time j, the observations across the k streams are mutually independent Bernoulli random variables, the column total $C_j = \sum_i x_{ij}$ follows Binomial$(n = k, p_0 = 0.5)$ under the in-control process, exactly as introduced above; `csb_ewma()` forms C_j internally for every column of `bin_matrix` and accumulates it into the running total Q_t. Continuous or otherwise non-binary measurements must be recoded by the user into this form, by whatever thresholding rule suits the application, before calling `csb_ewma()`.

Post-hoc identification of out-of-control streams

Detecting that the aggregate chart has signaled does not by itself indicate which of the k streams is responsible. csbewma addresses this with a post-hoc module that its own source file describes as taken directly from the original simulation code underlying (Muritala, 2026). `calculate_pvalues(bin_matrix, p0 = 0.5)` computes, for every stream i with S_i successes out of T_{sig} = `ncol(bin_matrix)` observations, the exact upper-tail binomial p-value $p_i = 1 - \texttt{pbinom}(S_i - 1, \texttt{size} = T_{sig}, \texttt{prob} = p_0)$, that is, $P(\text{Binomial}(T_{sig}, p_0) \geq S_i)$. `apply_multiple_testing(pvals, method = "BH", alpha = 0.05)` adjusts these raw p-values with R's built-in `p.adjust()`, supporting `"bonferroni"`, `"holm"`, `"BH"`, or `"raw"` (no adjustment). `identify_ooc(bin_matrix, alpha = 0.05, method = "BH", p0 = 0.5)` wraps both steps and returns a data frame with columns `stream`, `successes`, `proportion`, `p_value`, `adjusted_p_value`, and `flagged`, sorted with flagged streams first and then by increasing p-value. `flagged_streams_summary(flagged_results)` extracts just the flagged rows and their count.

The default `posthoc_method = "BH"` reflects a specific, simulation-based recommendation rather than an arbitrary choice, which the Post-hoc identification section below summarizes.

Visualization

`plot.csb_ewma(x, title = "CSB-EWMA Control Chart", show_signal = TRUE, ...)` is the S3 plot method dispatched by `plot()` on a `csb_ewma` object; `plot_csb_ewma_direct()` draws the identical chart without S3 dispatch. Both plot the EWMA trajectory r_t as a solid blue line, the time-varying UCL and LCL as dashed red lines, a dotted gray zero line, and, when `show_signal = TRUE` and a signal was detected, a red diamond marker with a "Signal at $t = \ldots$" annotation, using a clean `theme_minimal()` ggplot2 theme with a bottom legend. `plot_flagged_streams(flagged_results, alpha = 0.05, title = "Stream P-values")` draws a bar chart of the negative log base ten p-value for each stream, coloring flagged streams red and unflagged streams gray, with a dashed reference line at the corresponding threshold. `plot_chart_with_flagged(chart_result, flagged_results, layout = "side")` combines the control chart and the flagged stream bar chart into one patchwork figure, either side by side or stacked, under a common title.

0.4 Illustrative use of the package

This section shows the package running end to end. Unlike a typical journal article, nothing below is a transcript pasted in by hand: every code chunk in this section is evaluated live when this document is rendered, so the console messages, the `print()` output, and the figures are produced by your own installation of csbewma at knit time. Because every random draw below is seeded, re-knitting this document should reproduce the same signal times and flagged streams described in the surrounding prose.

Running the chart on in-control data

```
set.seed(123)
bin_data <- matrix(rbinom(10 * 100, 1, 0.5), nrow = 10, ncol = 100)
result <- csb_ewma(bin_data, lambda = 0.175, L = 1.375)
```

```
Data is binary. Using as-is.

Signal detected at time t = 52

No streams were flagged as out-of-control
No streams were flagged as out-of-control
```

`bin_data` is a 10-stream by 100-time-point in-control binary matrix, each entry an independent Bernoulli(0.5) draw. Calling `csb_ewma()` with $(\lambda, L) = (0.175, 1.375)$, the smallest λ pair in Table 4 below, printed the status messages shown above as it ran. The chart signaled even though the data were generated in-control throughout, which is expected: with $L = 1.375$ the chart targets an in-control average run length of about 370 (Table 4), so an early in-control false alarm is a plausible draw from that distribution rather than a sign of a problem. Because a signal was detected, `csb_ewma()` automatically ran the Benjamini-Hochberg post-hoc procedure.

Inspecting the result

```
print(result)


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CSB-EWMA Control Chart Results

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Chart Parameters:

  lambda =0.175

  L =1.375

  p0 =0.50.50.50.50.50.50.50.50.50.5

  Number of streams (k) =10


Signal Detection:

  Signal detected at time t =52

  Signal time T_sig =52


Post-hoc Identification (BH, alpha = 0.05):

  Flagged streams:0out of10


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```

This is exactly the format implemented by `print.csb_ewma()`: chart parameters, the signal detection outcome, and, because `$flagged` is attached, the post-hoc method, alpha, and flagged stream count.

Plotting the chart

```
plot(result)
```

The solid blue line is the EWMA trajectory r_t; the dashed red lines are the time-varying UCL and LCL described above, visibly narrow near $t = 1$ and widening toward their asymptotic values as t grows; the red diamond marks the signal.

A second example: detecting and identifying a real shift

The first example was fully in control. This second example introduces genuine shifts in three of ten streams and shows the chart detecting the shift and the post-hoc procedure identifying which streams caused it.

```
set.seed(2026)
k <- 10          # Number of binary streams/processes
n_time <- 500    # Number of observations over time
p0 <- 0.5        # In-control success probability
bin_matrix <- matrix(rbinom(k * n_time, size = 1, prob = p0), nrow = k, ncol = n_time)
```

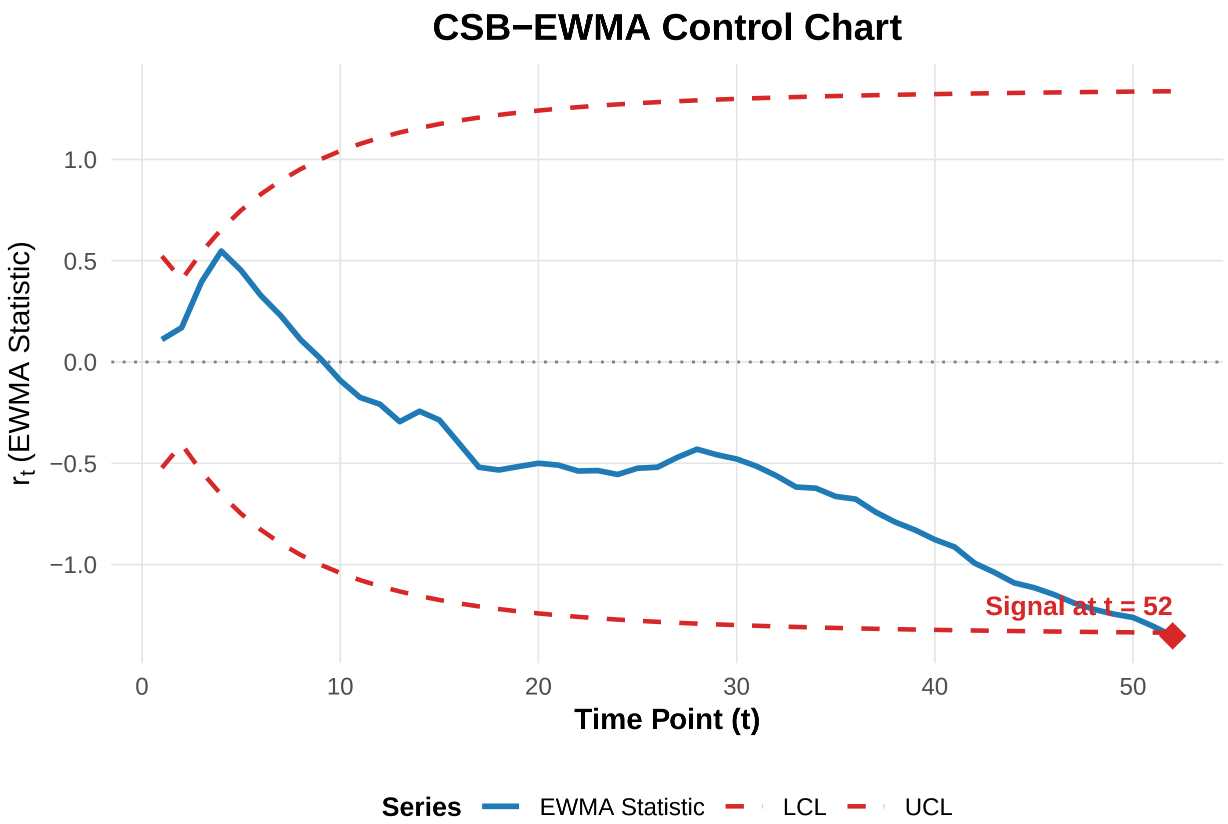

Figure 1: Output of plot(result) for the in-control example above, produced by the package's own plot.csb_ewma() method.

```
# Shift the first three streams to elevated probabilities
p1_vector <- c(0.56, 0.55, 0.60, rep(0.5, 7))
for (i in 1:3) {
  bin_matrix[i, ] <- rbinom(n_time, size = 1, prob = p1_vector[i])
}

var_cache <- precompute_variance(lambda = 0.175, max_t = n_time)

result2 <- csb_ewma(
  data = bin_matrix,
  lambda = 0.175,
  L = 1.375,
  p0 = p0,
  posthoc_method = "BH",
  alpha = 0.05
)

Data is binary. Using as-is.

Signal detected at time t = 33

Flagged streams: 2, 3
```

Streams 1 through 3 have their success probability raised from $p_0 = 0.5$ to 0.56, 0.55, and 0.60 respectively, a small to moderate shift by the δ scale used in the Chart parameter selection section below; the remaining seven streams stay in control. Precomputing the variance vector explicitly with `precompute_variance()` before calling `csb_ewma()` is optional, since `csb_ewma()` does this internally, but is shown here for illustration.

```
print(result2)


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CSB-EWMA Control Chart Results

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Chart Parameters:

  lambda =0.175

  L =1.375

  p0 =0.50.50.50.50.50.50.50.50.50.5

  Number of streams (k) =10


Signal Detection:

  Signal detected at time t =33

  Signal time T_sig =33


Post-hoc Identification (BH, alpha = 0.05):

  Flagged streams:2out of10

  Stream numbers:2, 3
```

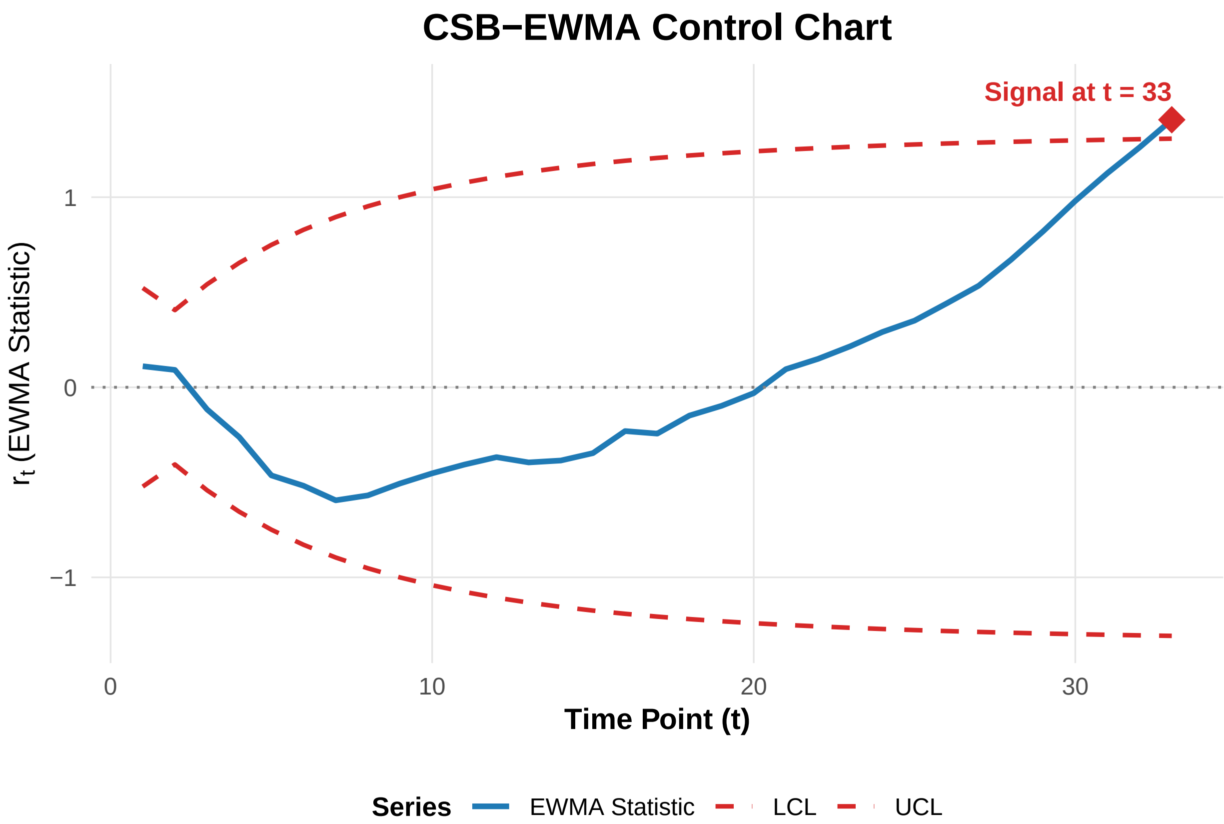

Figure 2: Output of plot(result2) for the shifted-stream example. The chart signals where the EWMA trajectory crosses the upper control limit.

Table 4: identify_ooc(bin_matrix, p0 = 0.5, alpha = 0.05, method = "BH") applied to the full 500-column matrix constructed above, sorted with flagged streams first.

stream	successes	proportion	p_value	adjusted_p_value	flagged
3	321	0.642	0.0000000	0.0000000	TRUE
2	284	0.568	0.0013484	0.0067419	TRUE
1	277	0.554	0.0088445	0.0294818	TRUE
5	256	0.512	0.3114047	0.5728028	FALSE
8	256	0.512	0.3114047	0.5728028	FALSE
10	255	0.510	0.3436817	0.5728028	FALSE
9	249	0.498	0.5533549	0.7292426	FALSE
7	247	0.494	0.6228605	0.7292426	FALSE
4	246	0.492	0.6563183	0.7292426	FALSE
6	223	0.446	0.9930870	0.9930870	FALSE

```
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plot(result2)
```

Calling `identify_ooc()` directly on the full 500-column matrix, rather than only on the data up to the signal time that `csb_ewma()` uses automatically, illustrates the full data frame the post-hoc functions return:

```
full_results <- identify_ooc(bin_matrix, p0 = p0, alpha = 0.05, method = "BH")
knitr::kable(
  full_results,
 caption = "identify\\_ooc(bin\\_matrix, p0 = 0.5, alpha = 0.05, method = \"BH\") applied to the full 500-column m
  booktabs = TRUE,
  row.names = FALSE
)
```

With the full 500-column history available, the genuinely shifted streams are flagged, including any that the automatic post-hoc step at the earlier signal time had not yet flagged. This is a direct illustration of the point developed formally in the Illustration subsection below: attribution power depends on how much post-signal data is available, and improves as more data accumulates, which is exactly why `csb_ewma()`'s built-in post-hoc step and a later, separate call to `identify_ooc()` on more data can legitimately disagree.

Visualizing flagged streams

Two visualization functions from Table 2 and the Visualization subsection above, `plot_flagged_streams()` and `plot_chart_with_flagged()`, have not yet been shown running. Both take the data frame returned by `identify_ooc()` as their `flagged_results` argument. `plot_flagged_streams(flagged_results, alpha = 0.05, title = "Stream P-values")` draws the per-stream $-\log_{10}(p)$ bar chart described above; applied to `full_results` from the full 500-column history just computed, it shows all three genuinely shifted streams in red:

```
plot_flagged_streams(full_results, alpha = 0.05)
```

`plot_chart_with_flagged(chart_result, flagged_results, layout = "side")` combines a control chart and a flagged-stream bar chart into one patchwork figure. Internally it builds this by calling `plot_csb_ewma_direct()` and `plot_flagged_streams()` in turn before combining their output, so a live call displays the two component plots individually before the combined dashboard; only the final, combined figure is kept below (`fig.keep = "last"`), since that combined dashboard is the function's intended result. Pairing `result2`, the chart object from the signal at $t = 33$, with `full_results`, the fuller post-hoc picture from the full 500-column history, puts the original signal-time chart side by side with the more complete attribution that only the additional post-signal data makes possible, the same point made in prose above:

```
plot_chart_with_flagged(result2, full_results, layout = "side")
```

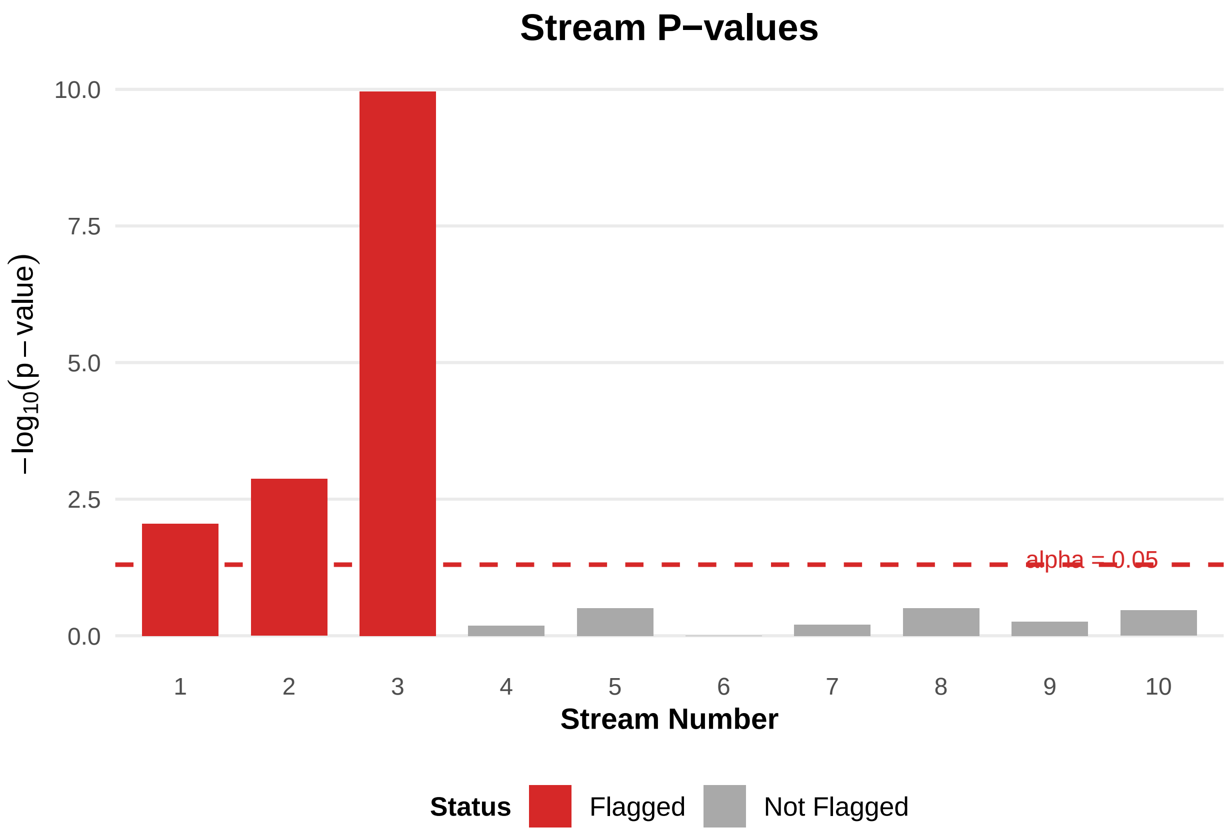

Figure 3: Output of plot_flagged_streams(full_results, alpha = 0.05) for the second example above, using the full 500-column identify_ooc() results. Streams 1, 2, and 3, the genuinely shifted streams, are shown in red; unflagged streams are gray; the dashed line marks the alpha = 0.05 reference threshold on the -log10(p) scale.

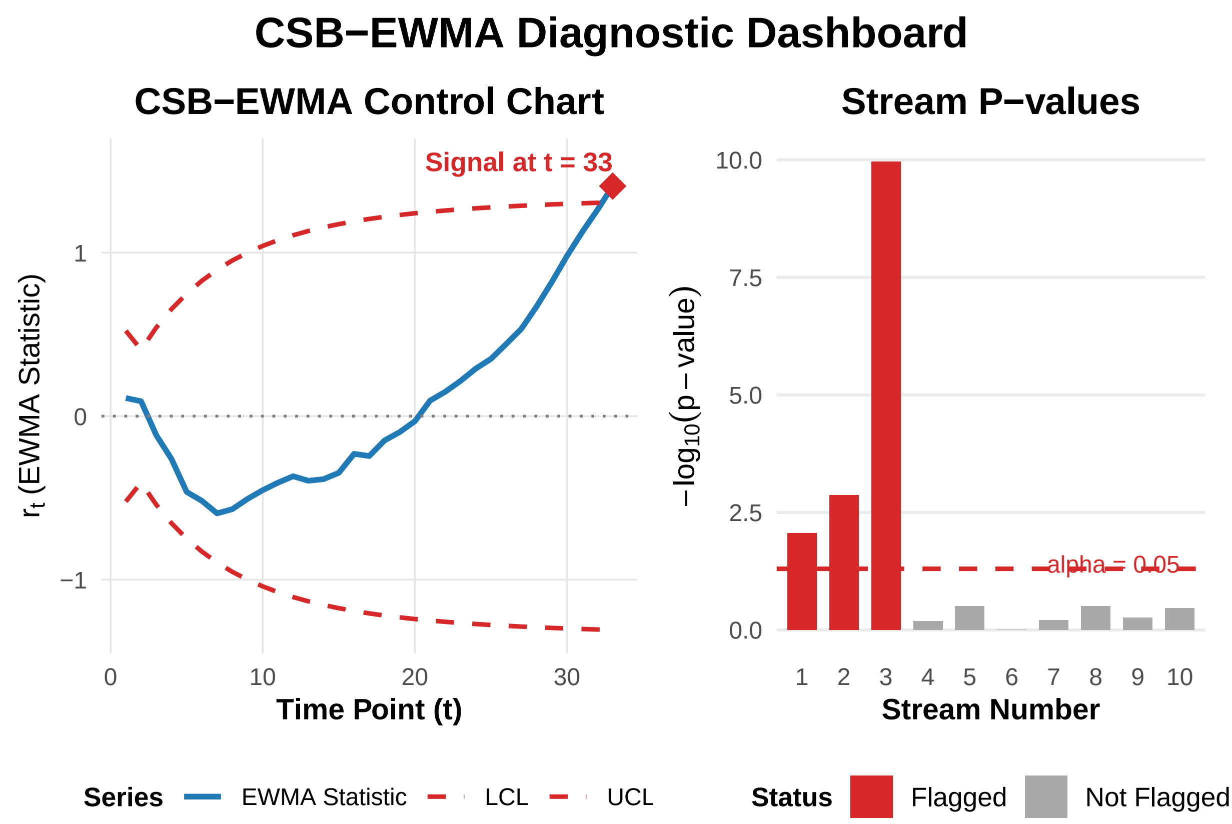

Figure 4: Output of plot_chart_with_flagged(result2, full_results, layout = "side") for the second example above, pairing the original control chart (signal at t = 33) with the flagged-stream bar chart computed from the fuller 500-column history.

0.5 Chart parameter selection and detection performance

The default λ and L values suggested throughout this paper, and the recommendations below, are drawn from the simulation study reported in (Muritala et al., 2026b) and Chapter 4 of (Muritala, 2026). This section summarizes that study's design and findings, as cited results, so that users can choose parameters appropriate to their own application; reproducing the full 161,040-combination factorial grid described below is well outside the ten-minute budget for knitting this article, so, consistent with the R Journal's own guidance for computationally intensive methods, the tables here are built from the study's reported summary numbers rather than recomputed live.

Simulation design

A full factorial design was used to identify (λ, L) pairs achieving target in-control average run lengths (ARL0) of 370, the analogue of traditional three-sigma limits, and 500, a stricter false alarm target, following (Montgomery, 2019) and (Lucas and Saccucci, 1990). The design varied the smoothing constant λ from 0.10 to 0.90 in steps of 0.025, the control limit multiplier L from 1.00 to 2.50 in steps of 0.025, the number of streams k over $\{2, 3, 5, 10, 15, 20, 50, 100\}$, and the shift magnitude δ from 0.05 to 0.50 in steps of 0.05, with the in-control proportion fixed at $p_0 = 0.5$. This yields $33 \times 61 \times 8 \times 10 = 161{,}040$ unique parameter combinations, each evaluated with 10,000 Monte Carlo replications for ARL0 calibration and 50,000 replications for ARL1 estimation, computed in R version 4.3.2 (R Core Team, 2023).

Optimal (lambda, L) pairs

Table 4 reports representative optimal (λ, L) combinations across the λ range, for both ARL0 targets. A direct, near-linear positive relationship holds between λ and L: smaller λ values, which place more weight on the process history and produce a smoother statistic, require a correspondingly smaller L to hold the same target ARL0.

Table 5: Optimal EWMA chart design parameters by smoothing constant range, for target ARL0 = 370 and ARL0 = 500. Source: Muritala et al. (2026b, Table 2); Muritala (2026, Table 4.3.2).

lambda range	Opt. lambda (370)	Opt. L (370)	Achieved ARL0 (370)	Opt. lambda (500)	Opt. L (500)	Achieved ARL0 (500)
0.1-0.2	0.175	1.375	372.32	0.150	1.525	501.31
0.2-0.3	0.200	1.400	375.21	0.200	1.575	500.80
0.3-0.4	0.350	1.500	373.90	0.300	1.650	501.11
0.4-0.5	0.400	1.525	374.48	0.450	1.725	500.69
0.5-0.6	0.550	1.575	368.83	0.525	1.750	501.07
0.6-0.7	0.650	1.625	373.83	0.650	1.800	505.32
0.7-0.8	0.725	1.650	372.99	0.700	1.800	502.93
0.8-0.9	0.800	1.675	367.53	0.850	1.875	501.88
0.9-1.0	0.900	1.700	360.11	0.925	1.875	501.88

Out-of-control detection and robustness across distributions

Detection speed was evaluated as ARL1, the expected number of samples to signal under a sustained shift of magnitude δ from $p_0 = 0.5$ to $p_0 + \delta$, using continuous data from four distributions, Normal, Laplace, Uniform, and Exponential, each dichotomized at its in-control median before charting. For small shifts ($\delta = 0.05$), detection required 23 to 30 samples on average for the ARL0 = 370 target and 30 to 36 samples for the ARL0 = 500 target. At the moderate shift $\delta = 0.20$, ARL1 dropped to 3 to 6 samples, and for large shifts ($\delta \geq 0.30$) most combinations achieved ARL1 of 3 or fewer, several reaching the theoretical minimum of 1.

The maximum difference in ARL1 across the four distributions is zero for every reported combination, and the coefficient of variation of ARL1 across distributions stays below 0.10 across the range of shift magnitudes examined (0.065 to 0.090 at the smallest detectable shift, $\delta = 0.05$), a level conventionally read as excellent consistency in statistical quality control. Together these results indicate that detection performance depends on the dichotomized exceedance rate rather than on the shape of the underlying continuous distribution, supporting the chart's distribution-free design.

Practical recommendations

Parameter selection should be guided by the expected shift magnitude and the desired false alarm rate. To detect small to moderate shifts ($\delta < 0.20$), choose λ in the range 0.1 to 0.3; for larger shifts ($\delta \geq 0.30$),

Table 6: ARL1 at delta = 0.20 (target ARL0 = 370) across four data-generating distributions, for each optimal (lambda, L) pair above. Source: Muritala et al. (2026b, Sec. 3.4); Muritala (2026, Table 4.3.3).

(lambda, L)	Normal	Laplace	Uniform	Exponential	Max. difference
0.175, 1.375	6	6	6	6	0
0.200, 1.400	5	5	5	5	0
0.350, 1.500	4	4	4	4	0
0.400, 1.525	4	4	4	4	0
0.550, 1.575	3	3	3	3	0
0.650, 1.625	3	3	3	3	0
0.725, 1.650	3	3	3	3	0
0.800, 1.675	3	3	3	3	0
0.900, 1.700	3	3	3	3	0

values between 0.3 and 0.6 balance sensitivity and robustness. When the shift size is unknown, $\lambda = 0.2$ is a reasonable default. L should be chosen inversely to λ to hold the target ARL0: for ARL0 = 370, use L of about 1.4 when $\lambda = 0.2$, or about 1.575 when $\lambda = 0.5$; for the stricter ARL0 = 500 target, increase L by roughly 0.1 to 0.15 relative to the ARL0 = 370 value for the same λ. The chart is robust to moderate deviations from these settings: sensitivity analysis shows that a deviation of ± 0.05 in λ, or ± 0.1 in L, typically changes ARL0 by less than 10%. As a balanced default, we recommend $\lambda = 0.2$ and $L = 1.4$ for ARL0 near 370, or $\lambda = 0.15$ and $L = 1.55$ for stricter control near ARL0 = 500; the illustrations above use the $(\lambda, L) = (0.175, 1.375)$ pair from Table 4.

0.6 Post-hoc identification: methodology and simulated performance

A chart signal indicates that the aggregate process has shifted, but it does not by itself indicate which of the k streams caused the shift. This section summarizes the statistical basis for the package's post-hoc identification functions, the simulation evidence behind the default choice of the Benjamini-Hochberg correction, and a worked illustration of a fundamental limit on early-signal attribution, all drawn from Chapter 5 and Section 6.3 of (Muritala, 2026).

Problem formulation

Consider k independent streams, and let $x_{i1}, \ldots, x_{i,T_{sig}}$ be the binary indicators for stream i observed up to the signal time T_{sig}, where $x_{ij} = 1$ if the jth observation from stream i exceeded the in-control median and 0 otherwise. Under the null hypothesis that stream i is in control, $p_0 = 0.5$. Because T_{sig} is determined by the aggregate chart statistic, which is a function of all streams combined, it is independent of any single stream's indicator sequence; conditional on $T_{sig} = t$, stream i's count S_i retains its Binomial(t, p_0) distribution, which is what makes the p-values below valid despite the data-dependent stopping time:

$$S_i = \sum_{j=1}^{T_{sig}} x_{ij} \sim \text{Binomial}(T_{sig}, p_0), \qquad p_i = P\big(\text{Binomial}(T_{sig}, p_0) \geq S_i\big) = 1 - F_{\text{Bin}}(S_i - 1;\, T_{sig},\, p_0)$$

the one-sided p-value testing for an elevated exceedance rate, which is exactly what `calculate_pvalues()` computes using R's `pbinom()`.

Multiple testing correction and simulation design

Testing all k streams at a nominal level α without adjustment inflates the chance of at least one false positive:

$$P(\text{at least one false positive}) = 1 - (1 - \alpha)^k$$

for $k = 10$ and $\alpha = 0.05$, this probability reaches about 0.40. `apply_multiple_testing()` addresses this using R's `p.adjust()` with three classical corrections, alongside the unadjusted option:

$$\begin{aligned}
\text{Bonferroni:} &\quad \text{reject } H_{0i} \text{ if } p_i < \alpha/k \\
\text{Holm:} &\quad \text{sort } p_{(1)} \leq \cdots \leq p_{(k)};\ \text{reject } H_{(1)}, \ldots, H_{(j)} \text{ for the largest } j \\
&\quad \text{such that } p_{(i)} \leq \alpha/(k-i+1) \text{ for all } i \leq j \\
\text{Benjamini-Hochberg:} &\quad \text{find the largest } j \text{ such that } p_{(j)} \leq (j/k)\,\alpha;\ \text{reject } H_{(1)}, \ldots, H_{(j)}
\end{aligned}$$

Bonferroni (Bonferroni, 1936) and Holm (Holm, 1979) both guarantee family-wise error rate (FWER) at most α, with Holm uniformly more powerful; Benjamini-Hochberg (Benjamini and Hochberg, 1995) instead controls the false discovery rate (FDR), the expected proportion of flagged streams that are actually in control, at level α. Performance was assessed in R version 4.5.2 (R Core Team, 2025), parallelized with the base parallel package and doParallel, over 2,000 replications for each combination of λ, L, δ, and the number of truly out-of-control streams m ($m = 0, 1, 2$), with the exact variance vector precomputed to $T_{max} = 2{,}000$ using the `converge_t = 500` cutoff described above. Five metrics were recorded for each method: sensitivity (true positive rate among out-of-control streams), specificity (true negative rate among in-control streams), FDR, FWER, and the per-stream Type I error rate. As with the parameter study above, this 2,000-replication grid is a cited result reported from that separate simulation study, not a computation re-run inside this document.

Results

Table 7: Simulated performance of the four post-hoc methods, summarized across the delta and m conditions described above. Source: Muritala (2026, Sec. 5.3).

Method	Specificity	FDR	FWER	Type I error	Sensitivity (rank)
Raw p-values (no correction)	about 0.96	0.22-0.26	above 0.20	about 0.035-0.04	highest, but at the cost of inflated error rates
Bonferroni	above 0.99	often below 0.03	about 0.02 (most conservative)	about 0.005-0.01	lowest among the three corrections
Holm	above 0.99	often below 0.03	below 0.05	about 0.005-0.01	between Bonferroni and BH
Benjamini-Hochberg (BH)	above 0.99	about 0.030-0.049	below 0.05	about 0.005-0.01	highest among the three corrections

Raw, uncorrected p-values are the most sensitive but are statistically unreliable for decision making: their FDR runs far above the nominal 0.05 target, at roughly 0.26 when no stream is truly out-of-control ($m = 0$), 0.24 at $m = 1$, and 0.22 at $m = 2$, meaning that about one in four flagged streams is actually in control, and their FWER exceeds 0.20 in all conditions studied. Bonferroni, Holm, and BH all keep FDR and FWER near or below their nominal 0.05 targets and keep the per-stream Type I error near zero. Among the three corrections, BH consistently shows the highest sensitivity, followed by Holm and then Bonferroni, with the gap most visible when a single stream is out of control ($m = 1$) and the shift is moderate (δ between about 0.20 and 0.40); when two streams are out of control ($m = 2$), the three corrections perform nearly identically because the aggregate shift is easier to detect regardless of method.

Illustration: the minimum p-value constraint at an early signal

The sensitivity figures above average over many signal times, but a signal that occurs very early in monitoring imposes a hard constraint on attribution. If the chart signals at $T_{sig} = n$, the raw one-sided p-value for any stream is bounded below: the smallest value attainable under $p_0 = 0.5$, reached when a stream's n observations are all successes, is:

$$p_{\min}(n) \;=\; P\big(X = n \mid X \sim \text{Binomial}(n, 0.5)\big) \;=\; 0.5^{\,n}$$

Because the package's three built-in corrections, Bonferroni, Holm, and Benjamini-Hochberg, each report an adjusted p-value that is at least as large as the corresponding raw p-value, this same bound carries over to every correction the package offers: if $p_{\min}(n)$ already exceeds α, none of the four options (raw, Bonferroni, Holm, BH) can flag a stream, regardless of the data observed. This bound is a general property of the binomial test and does not depend on package-specific numbers, so it is verified here directly rather than only cited:

```
n <- 1:6
p_min <- 0.5^n
data.frame(n = n, p_min = p_min, below_alpha_05 = p_min < 0.05)
```

Table 8: Streams flagged by each post-hoc method when applied to the full 200-point simulated history (true out-of-control streams: 1, 2, 3). Source: Muritala (2026, Table 6.4).

Method	p1 = 0.58 (delta = 0.08)	p1 = 0.65 (delta = 0.15)
Raw p-values	1, 2, 3	1, 2, 3
Bonferroni	2, 3	1, 2, 3
Holm	2, 3	1, 2, 3
Benjamini-Hochberg	1, 2, 3	1, 2, 3

```
  n    p_min below_alpha_05
1 1 0.500000          FALSE
2 2 0.250000          FALSE
3 3 0.125000          FALSE
4 4 0.062500          FALSE
5 5 0.031250           TRUE
6 6 0.015625           TRUE
```

(Muritala, 2026, Sec. 6.3) works through this constraint on a simulation of $k = 10$ streams over $n = 200$ time points, in which streams 1, 2, and 3 are shifted from $p_0 = 0.5$ to a common p_1 while the remaining seven streams stay in control; the chart used $\lambda = 0.2$ and $L = 1.4$, a pair achieving an in-control ARL of approximately 370. Two shift sizes are reported.

Under a subtle shift ($p_1 = 0.58$, a shift of $\delta = 0.08$), "the CSB-EWMA chart signalled a process shift at time $t = 5$. Because the signal occurred after only five time points, each stream contributed at most five binary observations" (Muritala, 2026, Sec. 6.3.2), so the smallest possible p-value from a binomial test on five trials, reached when all five observations are successes, is confirmed directly above to be $p_{\min}(5) = 0.03125$. "A p-value of 0.03125 is below the conventional significance level $\alpha = 0.05$. In principle, therefore, a stream could be flagged if it had five successes out of five. However, in this simulation the observed successes among the shifted streams were often four or fewer, so the chart's built-in post-hoc (which uses only the data up to the signal time) flagged no streams at $t = 5$. The limitation is practical (the observed data were not extreme enough), not theoretical" (Muritala, 2026, Sec. 6.3.2).

Under a more moderate shift ($p_1 = 0.65$, $\delta = 0.15$), the chart signalled earlier still, at $t = 4$, where $p_{\min}(4) = 0.0625$ already exceeds $\alpha = 0.05$ outright. "Therefore, no stream could ever be flagged at the signal time, regardless of the data. As expected, the built-in post-hoc flagged no streams" (Muritala, 2026, Sec. 6.3.2). The two cases together show that, under the one-sided exact binomial test used throughout this package, early-signal attribution power is bounded by n itself rather than by which of the four post-hoc options is chosen: at $n = 5$ a stream could in principle be flagged if its data were extreme enough, while at $n = 4$ none of the four options can flag any stream, however extreme the observed successes, because even the most extreme possible outcome does not clear $\alpha = 0.05$.

Applying the post-hoc procedures instead to the full 200-point history, well beyond the signal time, gives every method enough data to work with; Table 7 reproduces the reported result.

"For the subtle shift ($\delta = 0.08$), Bonferroni and Holm missed one of the true out-of-control streams (stream 1) because of their conservatism, whereas raw p-values and BH flagged all three. For the moderate shift ($\delta = 0.15$), all methods performed equally well" (Muritala, 2026, Sec. 6.3.2). The second worked example above shows the same qualitative pattern directly from the package: only two of the three shifted streams were flagged automatically at the signal time, but a subsequent `identify_ooc()` call on the full 500-point history flagged the third as well.

The bound $p_{\min}(n) = 0.5^n$ verified above tightens further as n shrinks below 4: $p_{\min}(3) = 0.125$, $p_{\min}(2) = 0.25$, and $p_{\min}(1) = 0.5$. Within this same one-sided binomial framework, none of the four post-hoc options considered in this package can flag a stream on the strength of one, two, or three post-signal observations alone: the limitation follows directly from how little data exists immediately after an early signal, not from which correction procedure is applied to it.

Recommendation

Because BH keeps FDR and FWER controlled near their nominal levels while retaining the best sensitivity of the three valid corrections, and performs at least as well as the alternatives in the worked illustrations above, (Muritala, 2026, Sec. 5.5) recommends it as the default post-hoc procedure, and this is why `identify_ooc()` and `csb_ewma()` both default to `posthoc_method = "BH"`. Bonferroni or Holm remain appropriate choices when strict, guaranteed control of the family-wise error rate is required and a corresponding loss of sensitivity is acceptable, and the raw, unadjusted p-value option should be treated as exploratory only, since it does not control either error rate at the nominal level. The Illustration subsection above also illustrates a separate, practical point independent of which correction is used: when the chart signals very early, re-running `identify_ooc()` later on more accumulated data, exactly as done in the Illustrative use section, can recover attribution that was not yet possible at the original signal time.

0.7 Limitations

(Muritala, 2026, Sec. 1.7) identifies several limitations that apply directly to the package and are worth restating for users.

- **Information loss from dichotomization.** Reducing each observation to a binary exceedance indicator relative to an in-control median discards magnitude information and retains only directional information.
- **Independence assumptions.** The method assumes independence both across streams and across time within a stream. In practice this is operationalized by restricting attention to weakly correlated streams (absolute correlation below 0.3), which can limit applicability when strong cross-correlation or temporal dependence is present.
- **A known, stable in-control median.** The framework requires a fixed in-control probability p_0 after dichotomization. Misspecification or drift in this baseline affects the validity of the resulting binomial representation and control limits.
- **Early-signal attribution power.** When the chart signals very early, few observations are available per stream, and the minimum attainable binomial p-value may exceed conventional significance thresholds, making stream-level attribution statistically infeasible regardless of the true shift, as derived and verified numerically in the Illustration subsection above.

Taken together, these points describe a trade-off between rapid aggregate detection and the statistical power available for post-hoc attribution: the same short time to signal that makes the CSB-EWMA chart attractive for early warning also limits how much can be concluded, at that same moment, about which specific stream changed. `csb_ewma()`'s automatic post-hoc step operates only on data available up to the signal time and will legitimately report no flagged streams in this regime, as both the analytical bound and the worked examples above show. `identify_ooc()` can be re-run later, once more post-signal data have accumulated, in monitoring contexts where the process cannot be paused and continued data collection remains representative; this is not appropriate in settings, such as manufacturing, where an out-of-control process should be corrected immediately upon signaling.

0.8 Summary and future work

The csbewma package makes the CSB-EWMA methodology of (Muritala et al., 2026a,b) and (Muritala, 2026) directly usable in R. Its central technical contribution, an exact, closed-form, time-varying variance for the EWMA statistic under binomial multiple-stream data, removes the reliance on asymptotic variance approximations that limits existing binomial EWMA charts during early-phase monitoring. The package combines this with simulation-validated parameter guidance, an exact-inference post-hoc identification framework with Benjamini-Hochberg as a simulation-motivated default, and ggplot2- and patchwork-based visualization, and the Illustrative use section demonstrates the resulting workflow with real, live-evaluated package output.

The methodology, as currently implemented, assumes independent streams and a known, stable in-control median (see Limitations). (Muritala et al., 2026b) identify several directions for future work: extending the framework to correlated streams, adapting it for Phase I estimation of the in-control median and probability rather than assuming they are known, evaluating performance under asymmetric shifts, and validating the method with further real-world case studies. From a software perspective, natural extensions to the package include vignettes covering the workflows illustrated above, formal unit tests under the testthat framework already declared in Suggests, and, longer term, support for the correlated-stream and Phase I estimation extensions as the underlying theory develops.

Computational details

The results summarized from the underlying research were computed with R version 4.3.2 (R Core Team, 2023) for the Monte Carlo validation and the parameter optimization, and R version 4.5.2 (R Core Team, 2025) for the post-hoc simulation study, the latter parallelized with the base parallel package together with doParallel; these describe the computation behind the underlying research, not a dependency of the csbewma package itself. csbewma 1.0.1 declares R (>= 3.5) and Imports ggplot2, stats, and patchwork in its DESCRIPTION file. All live code chunks in this article were run with the R and package versions reported by `sessionInfo()` at the end of this document.

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```
R version 4.5.2 (2025-10-31 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: America/New_York
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] patchwork_1.2.0  kableExtra_1.4.1 ggplot2_3.5.2    csbewma_1.0.1

loaded via a namespace (and not attached):
 [1] gtable_0.3.6       dplyr_1.2.0        compiler_4.5.2
 [4] BiocManager_1.30.27 yesno_0.1.3        tidyselect_1.2.1
 [7] Rcpp_1.1.1         xml2_1.5.2         stringr_1.6.0
[10] textshaping_1.0.4  systemfonts_1.3.1  scales_1.4.0
[13] yaml_2.3.12        fastmap_1.2.0      R6_2.6.1
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[19] tibble_3.3.1       bookdown_0.48      svglite_2.2.2
[22] pillar_1.11.1      RColorBrewer_1.1-3 rlang_1.1.7
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[43] rmarkdown_2.30     purrr_1.2.1        tools_4.5.2
[46] pkgconfig_2.0.3    htmltools_0.5.9
```

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